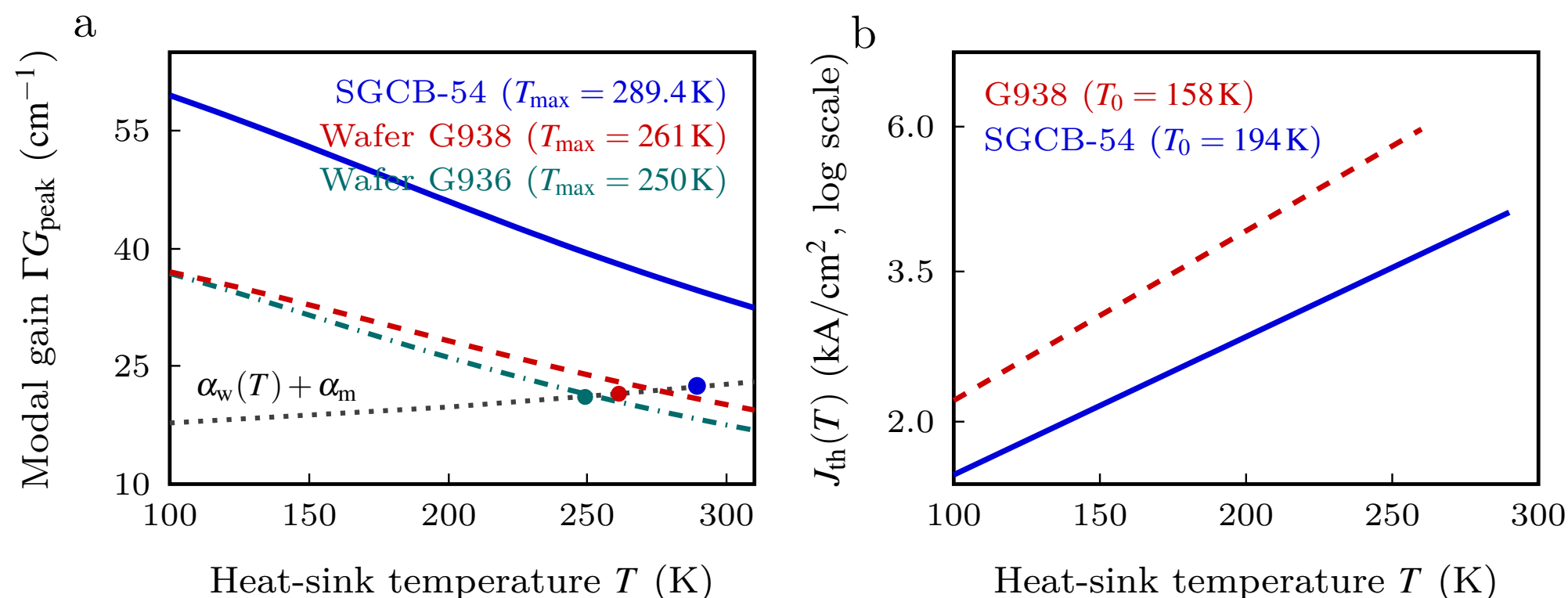
a
Modal gain ΓG_{peak} (cm^{-1})
55
40
25
10
100
150
200
250
300
Heat-sink temperature T (K)
SGCB-54 ($T_{\max} = 289.4$ K)
Wafer G938 ($T_{\max} = 261$ K)
Wafer G936 ($T_{\max} = 250$ K)
$\alpha_w(T) + \alpha_m$
b
$J_{\text{th}}(T)$ (kA/cm^2, log scale)
6.0
3.5
2.0
100
150
200
250
300
Heat-sink temperature T (K)
G938 ($T_0 = 158$ K)
SGCB-54 ($T_0 = 194$ K)